

TransitPulse

Live Bus Tracking and Fleet Management for Small Cities

Live bus tracker, driver app and dispatch center for Patiala City Transport

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Introduction

- Riders in small cities and tier-2 towns usually cannot tell when the next bus will arrive
- TransitPulse gives live GPS tracking, arrival times and free-seat counts
- Runs in any mobile browser: no app installation, low-bandwidth friendly



Commuter Tracker

Live buses, ETA, free seats



Driver Broadcaster

Location, speed, seat count



Dispatch Center

Routes, stops, passes, logs

Existing Work & Limitations



Existing practice

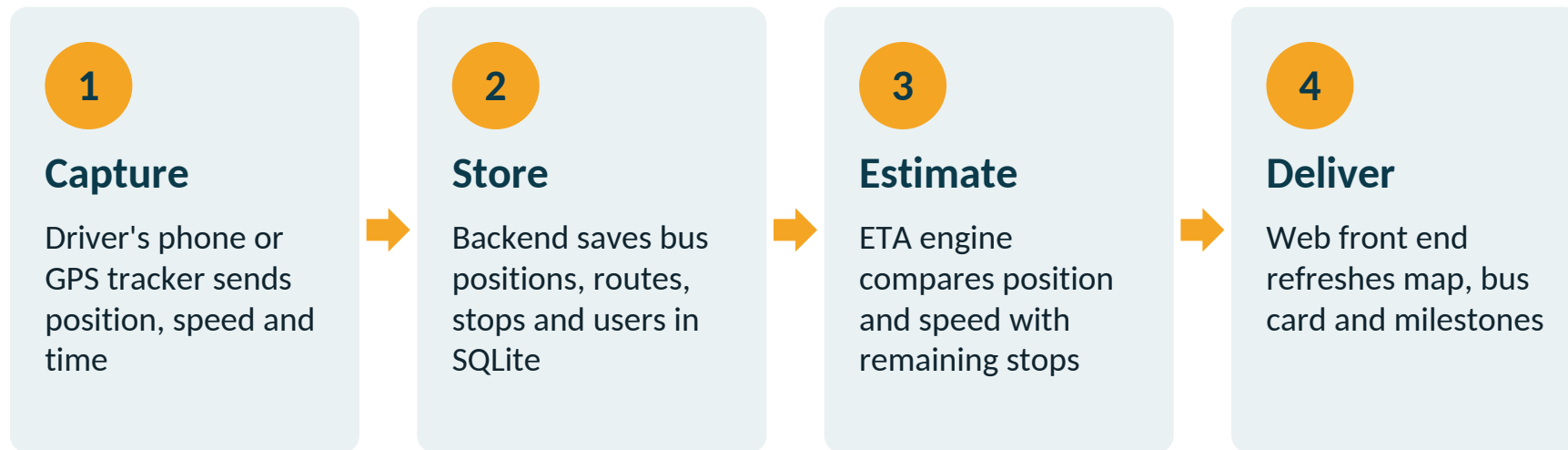
- Buses run with no real-time information for riders
- Unpredictable schedules: people wait for an unknown time or give up
- Many riders switch to private vehicles
- Authorities have no live view of the fleet



Limitations

- Over 60% of small-city commuters face delays (Urban Mobility India Report 2024)
- Buses bunch up: some overloaded, others nearly empty
- Falling ridership adds traffic and pollution
- Route planning relies on guesswork

Proposed Work & Methodology



Goals: live GPS tracking | clear arrival times at every stop | low-bandwidth operation | management tools for authorities

Novelty of the Project



Three parts, one platform

Tracker, driver app and dispatch center share one data source



Driver-reported crowding

Live seat map and vehicle status such as Flat Tyre reach riders and dispatchers



Geofenced stops

35 to 60 m geofences detect arrivals and drive trip milestones



Built for small towns

No app install, low bandwidth.

Real-Time Usage



At the bus stop

Rider checks the route and sees the live bus, ETA and free seats



In the control room

Fleet map refreshes every 3 seconds and shows reported breakdowns



In planning

Delay logs and the pass registry guide timetable changes

3 s

fleet map refresh

5

live route corridors

8

geofenced bus stops

HW & SW Requirements



Hardware

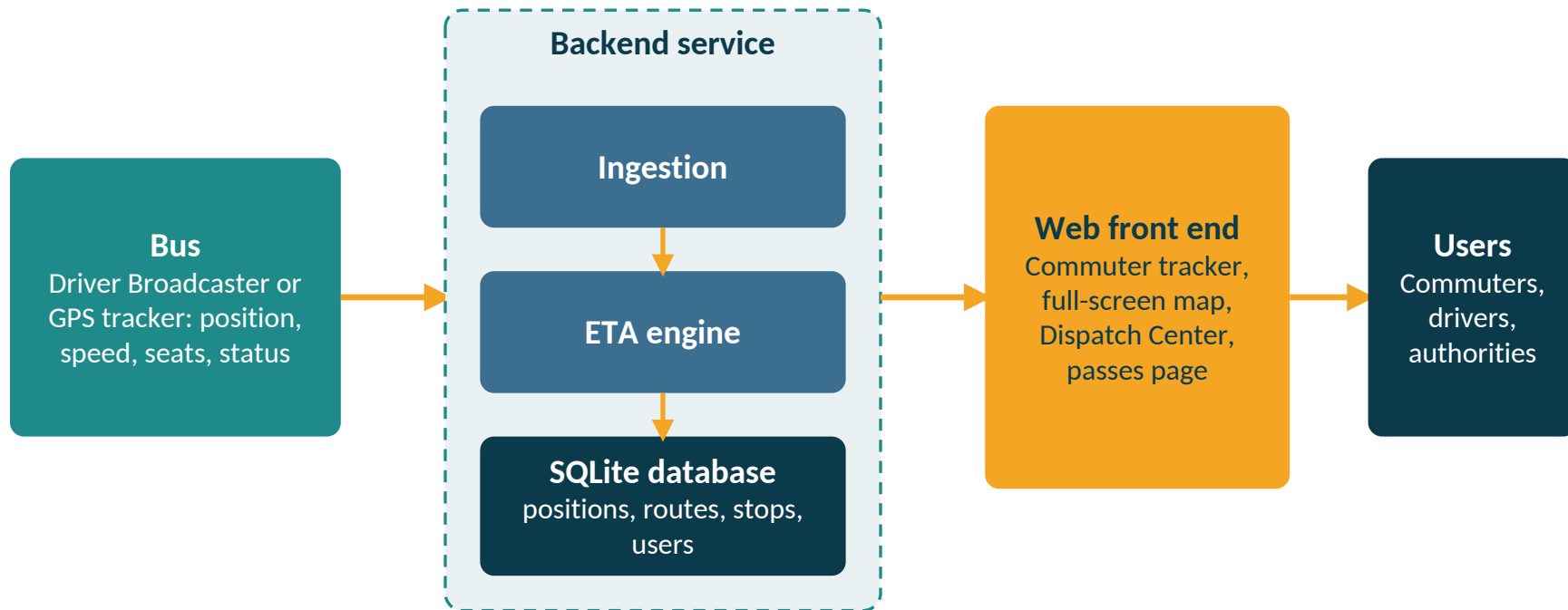
- Phone or computer with a web browser (commuters)
- GPS tracker or driver's smartphone on each bus
- Control-room computer for the Dispatch Center
- Server to host the backend service



Software

- Web browser with Leaflet interactive map
- Backend service for ingestion and ETA
- SQLite database
- Hosting on on render.com (free tier).

System Architecture



Capture > ingest and store > ETA > deliver. Operator tools use the same backend.

Module Description



Commuter Tracker

Route tabs, live map, ETA, free seats, milestones



Full-Screen Map

All buses on the network for control-room monitoring



Driver Broadcaster

Trip setup, live readings, 16-seat map, Start Trip



Civic Portal

Role-based sign-in and accounts backed by SQLite



Dispatch Center

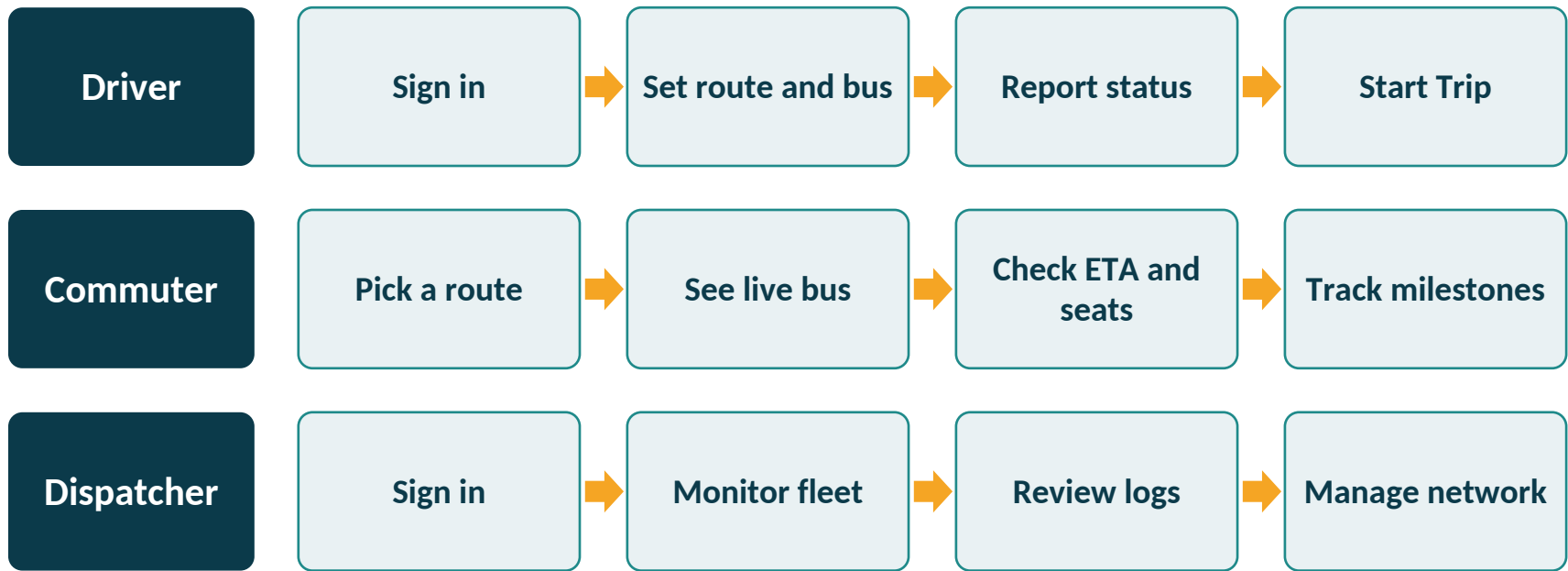
Fleet map, delay logs, routes, stops, users



Bus Passes

View and book passes; registry for authorities

Module Workflow



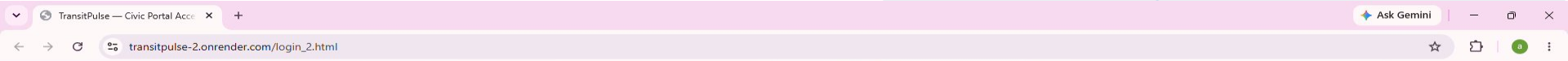
All three lanes read and write the same SQLite-backed data.

Implementation :

- Driver page posts position, speed, seats and status to the backend
- SQLite stores routes, stops, users, delay logs and passes
- ETA engine: distance remaining from position and recent speed
- Geofence check (35 to 60 m) updates trip milestones
- Leaflet map refreshes; fleet map every 3 seconds
- Role-based sign-in checked on the server

Demo Video

Live deployment
https://transitpulse-2.onrender.com/login_2.html





TransitPulse Civic Portal

Department of Transport • Patiala Municipal Transport

AUTHORIZED IDENTIFIER

SECURITY PIN / PASSWORD

Database Verified Access: Authenticate with credentials verified directly in the SQLite users database.

Snapshots: Commuter Tracker

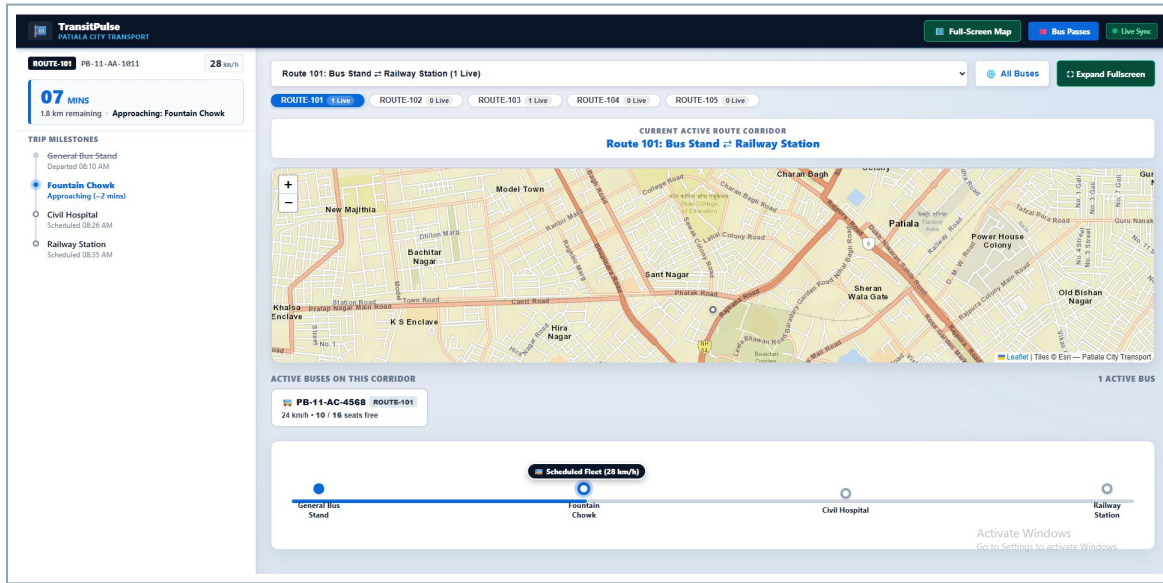


Figure 1: Live commuter tracker, Route 101 (Bus Stand to Railway Station)

Live map

Leaflet map , expandable to full screen

Active buses

Plate, speed, free seats (24 km/h, 10 of 16)

Trip status

ETA, 1.8 km remaining, next stop Fountain Chowk

Milestones

Stop-by-stop timeline and progress bar

Snapshots: Civic Portal

The screenshot shows the 'Sign In' page of the TransitPulse Civic Portal. At the top, the logo and title 'TransitPulse Civic Portal' are displayed, followed by the subtitle 'Department of Transport • Patiala Municipal Transport'. Below this, there are two buttons: 'Sign In' (highlighted in blue) and 'Create Account'. The 'AUTHORIZED IDENTIFIER' section contains a text input field with placeholder text 'e.g., admin_ops, adam_admin, driver_101'. The 'SECURITY PIN / PASSWORD' section contains a text input field with placeholder text '*****'. A blue box with white text states: 'Database Verified Access: Authenticate with credentials verified directly in the SQLite users database.' At the bottom, there is a large blue button labeled 'Verify & Enter Portal'.

Figure 2: Sign-in

The screenshot shows the 'Create Account' page of the TransitPulse Civic Portal. At the top, the logo and title 'TransitPulse Civic Portal' are displayed, followed by the subtitle 'Department of Transport • Patiala Municipal Transport'. Below this, there are two buttons: 'Sign In' and 'Create Account' (highlighted in blue). The 'CHOOSE USERNAME / IDENTIFIER' section contains a text input field with placeholder text 'e.g., adam_admin, harpreet_driver, rahul_commuti'. The 'CREATE SECURITY PIN / PASSWORD' section contains a text input field with placeholder text '*****'. The 'ACCOUNT TYPE' section contains a dropdown menu with 'Commuter Passenger' selected. A blue box with white text states: 'Self-Provisioning: New accounts are immediately committed to SQLite and available to the Admin command room.' At the bottom, there is a large green button labeled 'Complete Registration'.

Figure 3: Create account

The screenshot shows the 'Driver account type' page of the TransitPulse Civic Portal. At the top, the logo and title 'TransitPulse Civic Portal' are displayed, followed by the subtitle 'Department of Transport • Patiala Municipal Transport'. Below this, there are two buttons: 'Sign In' and 'Create Account' (highlighted in blue). The 'CHOOSE USERNAME / IDENTIFIER' section contains a text input field with placeholder text 'e.g., adam_admin, harpreet_driver, rahul_commuti'. The 'CREATE SECURITY PIN / PASSWORD' section contains a text input field with placeholder text '*****'. The 'ACCOUNT TYPE' section contains a dropdown menu with 'Onboard Bus Driver' selected. A blue box with white text states: 'Self-Provisioning: New accounts are immediately committed to SQLite and available to the Admin command room.' At the bottom, there is a large green button labeled 'Complete Registration'.

Figure 6: Driver account type

Credentials checked on the server against SQLite. Roles: Commuter Passenger, Onboard Bus Driver, Admin.

Snapshots: Driver App

TRANSITPULSE
Onboard Broadcaster

ASSIGNED CORRIDOR & DIRECTION
-- Choose Assigned Corridor --

VEHICLE REGISTRATION NO.
e.g., PB-11-AA-1011

TRAFFIC CONDITION
Normal

BUS / VEHICLE STATUS
Running Fine

SPEED
0.0 km/h

EMPTY SEATS
16 / 16

TRIP DURATION
00:00

LAST INGEST PING
-->--

LIVE COORDINATES
Lat: -- | Long: --

Figure 4: trip setup

Standby badge, route, plate, traffic and vehicle status, live readings

LIVE SEAT AVAILABILITY (TOP VIEW)

Empty Occupied

FRONT OF BUS			
S1	S2	S3	S4
S5	S6	S7	S8
S9	S10	S11	S12
S13	S14	S15	S16

REAR OF BUS

Reset All Seats to Empty

START TRIP

Figure 5: seat map

Green empty, red occupied; Start Trip begins broadcasting

TransitPulse Civic Portal
Department of Transport - Peel and Municipal Transport

Sign In Create Account

CHOOSE USERNAME / IDENTIFIER
e.g., adam_admin, harpreet_driver, rahul_commute

CREATE SECURITY PIN / PASSWORD

ACCOUNT TYPE
Onboard Bus Driver

Self-Provisioning: New accounts are immediately committed to S2Life and available to the Admin command room.

Complete Registration

Figure 6: driver sign-up with the Onboard Bus Driver type

Snapshots: Dispatch (1/3)

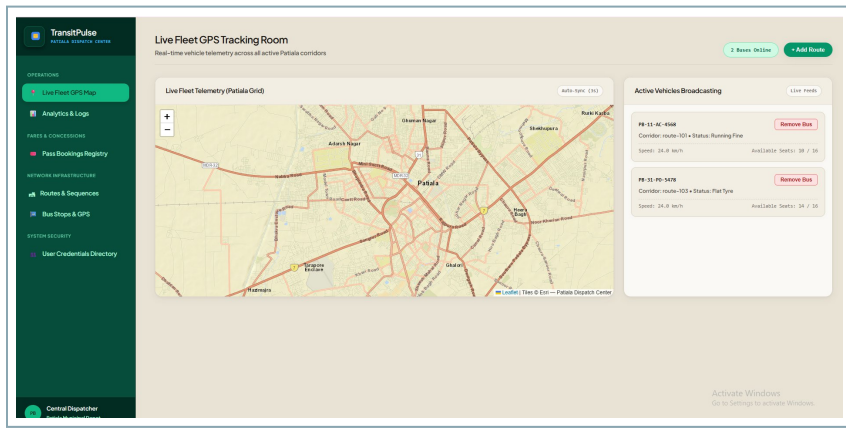


Figure 7: Live fleet GPS map

2 buses online; PB-31-PO-5478 reports Flat Tyre with 14 of 16 seats free

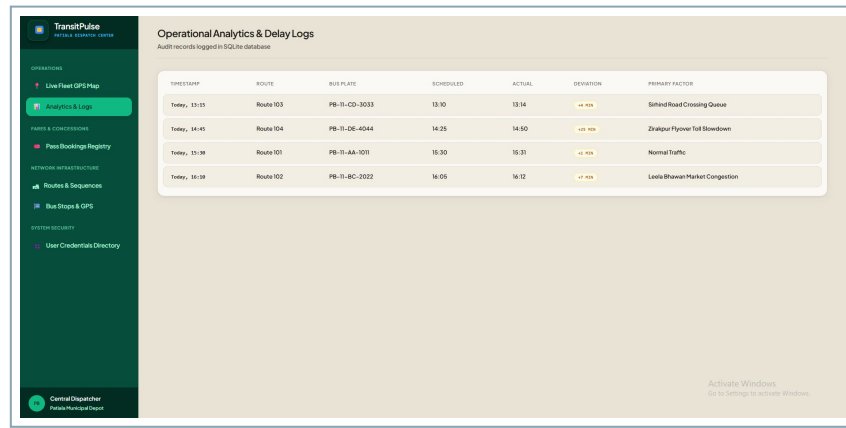


Figure 8: Analytics and delay logs

Delays from +1 min (Route 101) to +25 min (Route 104 toll slowdown)

Snapshots: Dispatch (2/3)

The screenshot shows the 'Municipal Concession Pass Issuance Registry' interface. It features a sidebar with navigation options like 'Live Fleet GPS Map', 'Analytics & Logs', 'Pass Bookings Registry', 'Routes & Sequences', 'Bus Stops & GPS', and 'User Credentials Directory'. The main content area displays a table of active passes with columns for Booking ID, Computer (Username), Pass Scheme, Shift Paid, Category, Validity Period, Date & Time Issued, and Status. Two active passes are listed, both for the 'Patiala City Hopper' scheme, valid for 24 hours, issued on 30/4/2023.

BOOKING ID	COMPUTER (USERNAME)	PASS SCHEME	SHIFT PAID	CATEGORY	VALIDITY PERIOD	DATE & TIME ISSUED	STATUS
886-4	scham_commuter	Patiala City Hopper	Rs 40	24 Hours (Unlimited)	24 Hours (Unlimited)	30/4/2023, 1:12:13 PM	Active
886-5	scham_commuter	Patiala City Hopper	Rs 40	24 Hours (Unlimited)	24 Hours (Unlimited)	30/4/2023, 2:42:51 PM	Active

Figure 9: Pass bookings registry

**Patiala City Hopper daily pass, Rs 40, 24 hours;
2 active passes**

The screenshot shows the 'Transit Network Corridors' interface. It features a sidebar with navigation options like 'Live Fleet GPS Map', 'Analytics & Logs', 'Pass Bookings Registry', 'Routes & Sequences', 'Bus Stops & GPS', and 'User Credentials Directory'. The main content area displays a table of active corridors with columns for Route ID, Corridor Title, Assigned Bus Plate, Default Speed, ETA, Distance, and Actions. Five corridors are listed, including routes 181, 182, 183, 184, and 185, each with a 'Delete Corridor' button.

ROUTE ID	CORRIDOR TITLE	ASSIGNED BUS PLATE	DEFAULT SPEED	ETA	DISTANCE	ACTIONS
route-181	Route 181: Bus Stand to Railway Station	PB-11-AA-1011	28km/h	07 mins	1.8km remaining	Delete Corridor
route-182	Route 182: Punjab University to Model Town	PB-11-BC-3022	32km/h	11 mins	3.4km remaining	Delete Corridor
route-183	Route 183: Tripuri to Urban Estate	PB-11-CD-3033	24km/h	05 mins	1.2km remaining	Delete Corridor
route-184	Route 184: Patiala to Chandigarh BRT-43 Express	PB-11-DE-4044	54km/h	22 mins	18.5km remaining	Delete Corridor
route-185	Route 185: Patiala to Ludhiana BRT Express	PB-11-EF-5055	48km/h	35 mins	29.0km remaining	Delete Corridor

Figure 10: Transit network corridors

**5 corridors with bus plate, default speed, ETA
and distance**

Snapshots: Dispatch (3/3)

TransitPulse
Bus Stops & GPS Geofences
Physical stops registered in the bus_stops table

+ Add New GPS Stop

STOP CODE	STOP NAME	LATITUDE	LONGITUDE	GEOFENCE RADIUS	ASSOCIATED ROUTES	ACTIONS
STP-01	Parkside Central Bus Stand	30.33981	76.38492	50 meters	Route 101, Route 104, Route 105	Delete
STP-02	Parkside Railway Station	30.32745	76.39802	40 meters	Route 101	Delete
STP-03	Purple University Gate	30.31631	76.40203	45 meters	Route 102	Delete
STP-04	Model Town Central	30.32489	76.37612	35 meters	Route 102	Delete
STP-05	Tripath Main Bazar	30.33023	76.38891	30 meters	Route 103	Delete
STP-06	Urban Estate Phase 2	30.3441	76.4332	40 meters	Route 103	Delete
STP-07	Chandigarh ISBT Sector 43	30.72191	76.74681	60 meters	Route 104	Delete
STP-08	Ludhiana Central ISBT	30.89942	75.85719	60 meters	Route 105	Delete

Activate Windows
Go to Settings to activate Windows.

Figure 11: Bus stops and GPS geofences

8 stops with 35 to 60 m geofences and serving routes

TransitPulse
Authorized User Accounts & Credentials Directory
Complete audit of all commuter, driver, and administrator accounts

+ Add Staff Account

DATABASE ID	USERNAME IDENTIFIER	SECURITY PIN / PASSWORD	SYSTEM ROLE	ACTIONS
#1	admin_ops	*****	ADMIN	Delete
#2	driver_driver	*****	DRIVER	Delete
#3	driver_driver	*****	DRIVER	Delete
#4	commuter_commuter	*****	COMMUTER	Delete
#5	rahal_commuter	*****	COMMUTER	Delete
#6	HANISH_DRIVER	*****	DRIVER	Delete
#7	admin_admin	*****	ADMIN	Delete
#8	a	*****	COMMUTER	Delete
#9	ab	*****	DRIVER	Delete
#10	abc	*****	ADMIN	Delete

Activate Windows
Go to Settings to activate Windows.

Figure 12: User accounts directory

10 accounts: 3 admins, 4 drivers, 3 commuters

Testing

Test	What was checked	Result
Live tracking	Route 101 shows live bus, speed, ETA and milestones	Observed
Data consistency	PB-11-AC-4568: 24 km/h, 10 of 16 seats on both tracker and fleet map	Consistent
Incident reporting	Flat Tyre status from the driver is visible to the dispatcher	Observed
Access control	Sign-in, role-based sign-up, new accounts in Admin directory	Observed
Hosted pages	Pages reachable; first load can take up to a minute on free tier	Observed

Results: Input & Output

INPUT

Driver picks a route, enters the plate, sets status Running Fine and taps Start Trip

Commuter opens Route 101

Driver of PB-31-PO-5478 reports Flat Tyre

New user registers as Onboard Bus Driver

OUTPUT

Bus appears on that route's tab with live speed and free seats (24 km/h, 10 of 16)

ETA, 1.8 km remaining, next stop Fountain Chowk and timed milestones

Dispatcher map lists route-103 bus as Flat Tyre, 14 of 16 seats free

Account saved to SQLite and listed in the Admin directory

Discussion

5

route corridors

8

geofenced stops

+25 min

worst logged delay

- Riders and operators read one consistent data feed
- Seat counts and breakdown status give crowding and incident awareness
- Delay causes reveal chronic bottlenecks for timetable changes
- ETA values are estimates: accuracy depends on GPS quality and traffic modeling

Conclusion & Future Scope

Conclusion

TransitPulse gives riders reliable real-time bus information through a lightweight web platform, gives drivers a simple way to broadcast trips and gives authorities a secure place to manage the network. With a stable feed, a well-tested ETA engine and a low-bandwidth focus, it can serve as a replicable model for other tier-2 towns.

Thank you

Future scope

- ML-based ETA prediction
- Crowding indicators from seat counts
- Punjabi and Hindi, voice and SMS alerts
- Arrival notifications for a chosen stop
- Digital payment for passes
- Analytics dashboard for authorities
- Open data API